The Cost of Passive Monetary Policy in Turkey

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Abstract

Turkey's transition from a successful post-2001 disinflation to renewed inflation under a prolonged passive monetary policy regime offers a natural setting for studying how the systematic stance of policy interacts with structural inflation persistence. We estimate a Markov-switching Taylor rule on monthly data and identify a sustained passive monetary policy episode running from February 2009 to May 2018, in which the central bank's inflation response coefficient falls to 0.67, well below the Taylor-principle threshold. A time-varying-parameter Kalman filter applied to 42 CPI sub-indices shows that inflation inertia is widespread, with sectors accounting for 62% of the consumption basket exhibiting above-mean persistence. Embedding both objects in a Bayesian Markov-switching structural VAR identified by sign-restricted monetary shocks, we construct a fully endogenous policy counterfactual that replaces the passive rule with a standalone active Taylor rule and re-simulates the entire system through the identified impulse responses. Had the central bank maintained an active stance over the episode, the price level would have risen by a cumulative 34.5 percentage points less (68% credible interval 27.9–40.4), with counterfactual inflation averaging 4.5% rather than the realized 8.3%. The counterfactual delivers only a partial disinflation rather than a return to target, a reflection of the structural inertia the same data reveal.

1 Introduction

Turkey entered the 2010s with single-digit inflation and exited the decade with inflation accelerating into the double digits and beyond. A natural question is how much of this trajectory reflects the systematic stance of monetary policy, how strongly the central bank leaned against inflation, and how much reflects structural features of price-setting that would have transmitted shocks into persistent inflation under any plausible policy. This paper develops an empirical framework to separate the two and to quantify what an alternative, rule-based policy would have delivered.

The period is unusually well-suited to identification because the change in the central bank's behaviour was driven less by the state of the economy than by external pressure on the conduct of policy. The shift toward an accommodative, growth-oriented stance over 2009–2018 coincides with a broader erosion of operational independence rather than with a textbook response to disinflation. This makes the regime change closer to an exogenous shift in the systematic rule than to an endogenous reaction, which justifies a counterfactual analysis.

The contribution of the paper is to bring together three elements previously studied separately for the Turkish case: a Markov-switching Taylor rule with the Davig and Leeper (2007) determinacy condition for formal regime dating, time-varying-parameter Kalman estimation of sectoral inflation persistence at the three-digit COICOP level, and a fully endogenous structural counterfactual. The counterfactual is the central methodological step. Prior Taylor-rule counterfactuals for Turkey have stopped at the rule-implied interest rate path, unable to extend to a structural model because the weak post-break rule produces indeterminacy in standard DSGE frameworks (Gürkaynak et al., 2015). The Markov-switching structural VAR used here, identified by sign restrictions on monetary shocks, propagates the alternative rule through the full seven-variable system without requiring a structurally determinate model under each regime.

We proceed in three stages. The first estimates a two-state Markov-switching Taylor rule to date the active and passive regimes and characterize the difference in the systematic inflation response between them. The second turns to the cross-section of prices: a time-varying-parameter Kalman filter recovers sector-specific inflation persistence across 42 CPI sub-indices, sectors are classified relative to the cross-sectional mean of the persistence parameter, and a principal-components step distills the cross-section into two latent factors, an inertial factor that loads on the high-persistence, domestically-driven sub-indices and an imported factor that loads on the exchange-rate- and commodity-sensitive ones. The third stage embeds these objects in a Bayesian Markov-switching structural VAR, identifies the monetary policy shock through sign restrictions, and constructs the central counterfactual of the paper.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 estimates the Markov-switching Taylor rule. Section 5 estimates sectoral inflation inertia and the latent factors. Section 6 develops the structural VAR and the policy counterfactual. Section 7 concludes.

2 Literature Review

This paper draws on three strands of literature: monetary policy rules under regime switching, inflation persistence in emerging markets, and structural VAR methodology for counterfactual evaluation.

2.1 Monetary Policy Rules and Regime Switching

The Taylor principle requires that the nominal policy rate respond more than one-for-one to inflation, ensuring that real interest rates rise when inflation accelerates (Taylor, 1993). Clarida et al. (2000) showed that the pre-Volcker Federal Reserve violated this condition ($\phi_\pi < 1$), while the post-Volcker regime satisfied it, linking Taylor principle compliance to macroeconomic stability. Davig and Leeper (2007) extended the analysis to regime-switching environments, establishing a generalized determinacy condition: the economy is determinate if the ergodic-probability-weighted average inflation response exceeds unity, even if the passive regime alone would produce indeterminacy. The present paper applies this condition to Turkey, using the first violation of the Davig-Leeper threshold as the formal intervention date for the counterfactual.

The broader Markov-switching VAR framework follows Sims and Zha (2006), who estimate regime changes in both the systematic policy component and shock volatility for the United States; their Bayesian estimation strategy informs the MS-SVAR estimated below. For Turkey, Yağcıbaşı and Yıldırım (2019) estimate a Markov-switching Taylor rule and find active and passive regimes consistent with post-2001 reform followed by an erosion of monetary discipline, but do not connect the regime parameters to sectoral inflation dynamics or to a counterfactual framework.

2.2 Turkey's Monetary Policy Experiment

The structural break in the CBRT's Taylor rule around 2009 was documented by Gürkaynak et al. (2015), whose structural-break tests locate the break in 2009 in all formulations of the rule and whose post-break inflation response coefficient ($\phi_\pi \approx 0.50$–0.64) is closely aligned with the passive-regime estimate in the Markov-switching specification used here ($\phi_\pi^P = 0.67$). Gürkaynak et al. (2023) extend the analysis and provide a detailed account of the monetary policy experiment over the longer horizon, documenting the drift below unity around 2010–2011 and arguing that the shift was driven by political pressure on the conduct of policy rather than by fiscal dominance or the underlying state of the economy.

Three features of this body of work are directly relevant. First, the finding that the policy shift was exogenous, a change in the systematic rule rather than an optimal response to changing conditions, supports the identifying assumption underlying the structural counterfactual in Section 6. Second, the dating broadly agrees with the February 2009 regime switch identified here: the structural-break tests of Gürkaynak et al. (2015) point directly to 2009, while Gürkaynak et al. (2023)'s continuous TVP estimation locates the drift below unity around 2010–2011, slightly later than the discrete Markov-switching dating. The three studies also use different policy rate measures, the overnight borrowing rate in the present paper, the one-week TRlibor rate in Gürkaynak et al. (2015), and the weighted average cost of CBRT funding in Gürkaynak et al. (2023), but the qualitative dating is robust to this choice. Third, Gürkaynak et al. (2015) implement a simple Taylor-rule counterfactual, estimating the interest rate the pre-break rule would have prescribed in the post-break period, and find the actual policy rate was about seven percentage points below the counterfactual; they were unable, however, to extend this to a structural counterfactual because the weak post-break rule produces indeterminacy in standard DSGE models. The Markov-switching framework with the Davig and Leeper (2007) generalized determinacy condition, employed here, accommodates indeterminacy at the level of a single regime while remaining tractable for the system-wide counterfactual of Section 6.

2.3 Inflation Persistence and Sectoral Dynamics

Bilici and Çekin (2020) estimate time-varying inflation persistence for Turkey using an AR(2) model of headline CPI, finding that persistence declined after institutional reforms and began rising around 2016. The present paper's backward-indexation weight $\alpha_{i,t}$ is a related but distinct quantity: estimated within a state-space model with controls for exchange rates, commodity prices, and the output gap, it isolates the backward-looking pricing component after accounting for contemporaneous cost-push factors. The CPI-weighted aggregate of these estimates shows re-embedding beginning several years before the aggregate persistence rise documented by Bilici and Çekin (2020).

Sakarya et al. (2025) examine inflation inertia using an augmented Phillips Curve with a total connectedness index, finding that inertia increased markedly after 2018 through stronger expectations and intensified price interconnectedness. Their conclusion explicitly identifies sectoral decomposition as a direction for future research. The present paper responds by estimating individual backward-indexation weights for each of the 42 three-digit COICOP sub-indices, classifying them into structurally distinct groups, extracting common factors via PCA, and embedding the resulting factors in a regime-switching SVAR counterfactual.

2.4 Structural VAR and Factor Methods

The third stage employs a Bayesian MS-SVAR identified through sign restrictions following Uhlig (2005), with commodity prices included following Sims (1992) to resolve the price puzzle. Sign restrictions identify a set of admissible structural models rather than a point estimate; subsequent work has refined the approach with elasticity bounds (Kilian and Murphy, 2012), penalty-function methods for jointly identifying multiple shocks (Mountford and Uhlig, 2009), and narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018). The present paper adopts the magnitude-bound logic of Kilian and Murphy (2012) together with a cumulative post-filter on the price response. The Bayesian estimation uses Minnesota priors (Doan et al., 1984) with equation-by-equation Gibbs sampling as in Kadiyala and Karlsson (1997).

The use of PCA-extracted factors as VAR variables follows the factor-augmented VAR methodology of Bernanke et al. (2005), with the theoretical foundations for principal components as consistent estimators of latent common factors established by Stock and Watson (2002); Bai and Ng (2002) provide formal criteria for selecting the number of factors in larger panels, though the present application takes one factor from each of two pre-defined sectoral groups by construction. The present paper adapts this approach by extracting factors not from a generic macroeconomic panel but from two economically motivated groups of CPI sub-indices, producing an inertial factor ($\hat{F}_t^{iner}$) and an imported shock factor ($\hat{F}_t^{imp}$) with clear economic interpretation.

2.5 Contribution

The present paper combines three elements: (i) a Markov-Switching Taylor Rule with the Davig and Leeper (2007) determinacy condition for formal regime dating, complementing Gürkaynak et al. (2023) and Yağcıbaşı and Yıldırım (2019); (ii) TVP-Kalman estimation of individual sub-index backward-indexation weights, extending the aggregate analysis of Bilici and Çekin (2020) to the three-digit COICOP level; and (iii) a Bayesian MS-SVAR counterfactual that embeds the regime dates, Taylor Rule parameters, and sectoral factors in a unified framework. Stages 1 and 2 are estimated independently; both provide inputs to Stage 3.

3 Data: Sources, Transformations, and Summary Statistics

The analysis uses monthly data from January 2003 to December 2025. All series are sourced from the CBRT Electronic Data Delivery System (EVDS), the Turkish Statistical Institute (TÜİK), and the Federal Reserve Economic Data (FRED) service.

3.1 CPI Sub-Indices

The CPI sub-index data are drawn from TÜİK's archived Consumer Price Index (base year 2003=100). This classification organizes household expenditures into 12 major groups at the two-digit level and 44 three-digit sub-groups; the international COICOP sub-group 4.2 (imputed rentals for housing) is not compiled by TÜİK and is absent from the Turkish classification, so these 44 are the sub-groups actually reported for Turkey. From them we exclude two series: J103 (post-secondary non-tertiary education), terminated in December 2015 and dropped for incomplete coverage, and J126 (financial services n.e.c.), which is ill suited for the TVP-Kalman specification. The final panel comprises 42 sub-indices.

CPI basket weights at the three-digit level are constructed from TÜİK's 2026 item-level weight publication. Since the weights enter only the descriptive aggregate α trajectory and basket composition statistics (not the Kalman filter, PCA, or SVAR) any approximation from using end-of-sample weights affects only the Stage 2 descriptive figures.

3.2 Macroeconomic Variables

Policy rate. The overnight borrowing rate of the CBRT (monthly average). The Markov-Switching framework captures discrete regime changes in the rule's behavior and is qualitatively robust to the choice of rate measure.

Exchange rate. The USD/TRY nominal exchange rate (monthly average, EVDS) in month-on-month log-differences ($\Delta e_t = \ln(e_t/e_{t-1}) \times 100$).

Commodity prices. The IMF All Commodity Price Index (FRED series `PALLFNFINDEXM`) in month-on-month log-differences. This variable enters both the TVP-Kalman measurement equation as a control for imported cost shocks and the SVAR as a separate variable, following Sims (1992), to resolve the price puzzle.

Output gap. Derived from the seasonally and calendar adjusted Industrial Production Index (TÜİK, base year 2021=100) via the Hodrick-Prescott filter (Hodrick and Prescott, 1997) applied to the log of the index, with the cyclical component expressed in percentage terms. The smoothing parameter is set to $\lambda = 14{,}400$, the standard value for monthly data under the frequency adjustment of Ravn and Uhlig (2002).

Headline CPI. Month-on-month log-difference of TÜİK's headline CPI index, multiplied by 100.

3.3 Variable Transformations for the SVAR

The seven-variable VAR is estimated on:

1. **Output gap:** HP-filtered log IPI $\times$ 100.

2. **ΔCommodity:** Month-on-month log-difference of IMF All Commodity Price Index $\times$ 100.

3. **ΔExchange rate:** Month-on-month log-difference of USD/TRY $\times$ 100.

4. **$\hat{F}_t^{imp}$:** First principal component of monthly inflation rates of the low-inertia sub-indices (30% of group variance explained).

5. **$\hat{F}_t^{iner}$:** First principal component of monthly inflation rates of the high-inertia sub-indices (51% of group variance explained).

6. **π_{CPI}:** Month-on-month log-difference of headline CPI $\times$ 100.

7. **Policy rate:** Overnight borrowing rate (% per annum).

The ordering reflects standard assumptions: slow-moving real variables first, financial and price variables in the middle, the policy instrument last. Commodity prices are ordered second as largely exogenous to Turkey's domestic economy.

3.4 Sample Periods

The CPI sub-index data span January 2003 to December 2025 (276 months). The TVP-Kalman filters are estimated on this full sample. The SVAR is estimated on the 273 monthly observations available after differencing and the two autoregressive lags. The counterfactual intervention window is February 2009 to May 2018 (112 months), corresponding to the primary passive regime episode identified in Stage 1.

4 Markov-Switching Taylor Rule Estimation

This section estimates a Markov-Switching Taylor Rule for the CBRT and applies the Davig and Leeper (2007) generalized determinacy condition to formally date the passive monetary policy regime.

4.1 Model Specification

The CBRT's policy rule is modeled as a two-regime Markov-Switching Taylor Rule:

$$i_t = \alpha^{(s_t)} + \phi_\pi^{(s_t)}\pi_{t-1} + \phi_y^{(s_t)}y_t + \varepsilon_t^{(s_t)}, \quad \varepsilon_t^{(s_t)} \sim N(0, \sigma^2_{(s_t)}) \tag{1}$$

where i_t is the overnight borrowing rate (% per annum), π_{t-1} is lagged month-on-month CPI inflation, y_t is the output gap, and $s_t \in \{0, 1\}$ is the unobserved regime indicator governed by a first-order Markov chain with transition matrix

$$P = \begin{pmatrix} p_{00} & 1 - p_{00} \\ 1 - p_{11} & p_{11} \end{pmatrix} \tag{2}$$

Lagged inflation π_{t-1} is used rather than contemporaneous inflation to reflect the Monetary Policy Committee's information set at each rate decision.

4.2 Estimation

The model is estimated by maximum likelihood via the Hamilton filter (Hamilton, 1989) on 275 monthly observations. Kim-Nelson smoothing provides smoothed regime probabilities $P(s_t = j \mid \mathcal{I}_T)$. The regime with the higher inflation response coefficient is identified as active.

4.3 Parameter Estimates

Table 1 reports the estimated parameters.

Table 1: Markov-Switching Taylor Rule: Parameter Estimates

	Passive Regime ($s_t = 0$)		Active Regime ($s_t = 1$)	
Parameter	Estimate	Std. Error	Estimate	Std. Error
α (intercept)	5.977	0.222	20.040	1.255
ϕ_π (inflation response)	0.674	0.101	2.944	0.559
ϕ_y (output gap response)	-0.103	0.036	-0.799	0.302
σ^2 (error variance)	4.398	0.585	97.199	11.994
Model fit				
Log-likelihood		-827.36		
AIC		1674.72		
BIC		1710.88		

The active regime's inflation response of $\phi_\pi^A = 2.94$ (s.e. 0.56) is well above the Taylor principle threshold, while the passive regime's $\phi_\pi^P = 0.67$ (s.e. 0.10) is below unity, indicating that the CBRT allowed real interest rates to fall when inflation rose. Both are significant at the 1% level.

The output gap coefficient is negative and statistically significant in both regimes ($\phi_y^P = -0.103$, $p = 0.005$; $\phi_y^A = -0.799$, $p = 0.008$). The negative sign runs contrary to a conventional stabilizing rule, under which the policy rate rises when output exceeds potential. It most plausibly reflects within-month simultaneity between the policy rate and industrial production rather than a deliberate stabilization response.

The error variance is substantially larger in the active regime ($\sigma_A^2 = 97.2$ vs. $\sigma_P^2 = 4.4$), reflecting crisis tightening episodes with large, discrete rate adjustments that the linear specification cannot fully capture.

Table 2 reports the transition dynamics.

Table 2: Regime Transition Dynamics

Parameter	Estimate
P(stay passive | passive)	0.976
P(stay active | active)	0.981
P(passive $\rightarrow$ active)	0.025
P(active $\rightarrow$ passive)	0.019
Ergodic probability, active	0.560
Ergodic probability, passive	0.440
Expected duration, active	51.9 months
Expected duration, passive	40.8 months

Both regimes are highly persistent, with stay probabilities exceeding 0.97. The ergodic distribution assigns 56% of time to the active regime and 44% to the passive regime.

4.4 Regime Dating

The smoothed regime probabilities identify three passive episodes:

- February 2009 to May 2018 (112 months), the primary episode
- December 2019 to October 2020 (11 months)
- January 2022 to May 2023 (17 months)

The primary episode begins in February 2009, shortly after the CBRT's aggressive rate cuts during the Global Financial Crisis. That recession was the most severe Turkey had experienced in decades, triggered by a collapse in external demand: real GDP fell by 4.9% in 2009, and the CBRT cut its policy rate by more than 1,000 basis points from October 2008 as a large negative output gap opened, a counter-cyclical response made possible by the sounder post-2001 macroeconomic framework (Rawdanowicz, 2010). These initial cuts were thus a defensible response to a downturn of that magnitude; the passivity the regime classification captures is the CBRT's subsequent failure to return rates to Taylor-principle-consistent levels as the economy recovered. The opening months of the episode therefore coincide with a period of warranted crisis easing. This dating is approximately two years earlier than the deliberate corridor framework described by Gürkaynak et al. (2023), whose continuous TVP estimation shows $\phi_{\pi,t}$ drifting below unity around 2010–2011; both methods agree that the Taylor principle was violated by 2010 at the latest.

The contrast between the contemporaneous and the retrospective assessment sharpens the interpretation of the regime. Writing in real time at the end of 2010, the OECD still read the post-crisis cuts as temporary stimulus and recommended that the normalization of policy rates begin before the end of that year, warning against a repeat of the 2006–07 episode of unanchored expectations (Rawdanowicz, 2010). That normalization never arrived: the response coefficient instead drifted below unity over precisely this window, an outcome Gürkaynak et al. (2023) attribute to political pressure on the conduct of policy. The passive regime dated here is exactly this gap: between the prompt unwinding of crisis stimulus that the policy framework of the time was expected to deliver and the accommodation that persisted in its place. That the accommodation departed from the contemporaneous benchmark, rather than tracking it, supports reading the regime change as a shift in the systematic rule rather than as an optimal response to conditions. This is the premise on which the counterfactual of Section 6 rests.

A concrete illustration of the entrenched passive stance comes from late 2013 and early 2014. The Federal Reserve's taper signal triggered capital outflows and a sharp lira depreciation, but the CBRT initially declined to raise rates at its scheduled December 2013 meeting; the lira depreciated further, and only after a substantial additional loss of value did the CBRT convene an emergency meeting in late January 2014, raising the overnight borrowing rate from approximately 3.5% to 8% in a single step (Gürkaynak et al., 2015). The borrowing rate remained near that elevated level through the rest of the passive episode, while the CBRT eased its one-week repo rate (the main funding instrument at the time, distinct from the corridor floor) through 2014 and 2015 under political pressure (Gürkaynak et al., 2023). What the Markov-switching specification identifies as passive is the systematic responsiveness of policy to inflation: the smoothed regime probabilities continue to classify the post-January 2014 period as passive, and the regime-wide estimate $\phi_\pi^P = 0.67$ pools over the whole episode without indicating a switch. The model thus reads the January 2014 hike as a level shift within the passive regime rather than as a regime switch, an emergency intervention that raised the corridor floor without restoring the systematic responsiveness of policy, leaving the underlying rule passive throughout.

The primary episode ends in May 2018, coinciding with the emergency tightening during the lira crisis. The two shorter episodes correspond to pandemic-era easing (December 2019 – October 2020) and the unorthodox rate-cutting experiment (January 2022 – May 2023). The counterfactual in Section 6 focuses on the primary episode.

4.5 The Davig-Leeper Determinacy Condition

The generalized determinacy condition requires that the ergodic-probability-weighted average inflation response exceed unity:

$$\bar{\phi}_\pi = \omega_A \, \phi_\pi^A + \omega_P \, \phi_\pi^P > 1 \tag{3}$$

Evaluated at the estimated ergodic probabilities:

$$\bar{\phi}_\pi = 0.560 \times 2.944 + 0.440 \times 0.674 = 1.945 \tag{4}$$

The unconditional determinacy condition is satisfied ($\bar{\phi}_\pi = 1.95 > 1$). However, substituting smoothed regime probabilities at each date produces a time-varying $\bar{\phi}_\pi(t)$ that drops to approximately 0.67 during the primary passive episode, deep in the indeterminacy region. The condition first falls below 1.0 and remains there for more than six consecutive months beginning in February 2009. This date serves as the formal intervention point for the counterfactual.

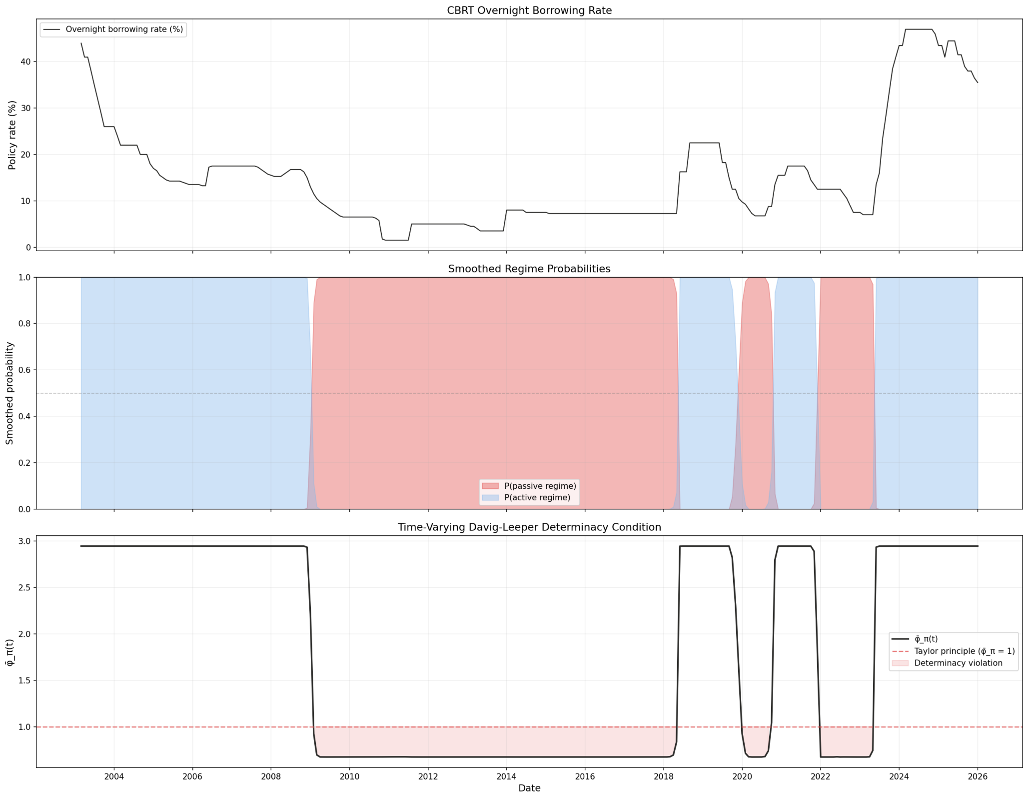

Figure 1: Smoothed regime probabilities and the time-varying Davig-Leeper determinacy parameter $\bar{\phi}_\pi(t)$. The dashed line at $\bar{\phi}_\pi = 1$ marks the determinacy threshold.

4.6 Robustness

As a robustness check, the model is re-estimated using an AR(1) one-step-ahead inflation forecast. The coefficient estimates shift (the less responsive regime produces $\phi_\pi = 1.29$ (compared to 0.67 in the baseline), the more responsive $\phi_\pi = 6.52$ (compared to 2.94)) a well-documented sensitivity of Taylor Rule estimation to the inflation measure (Clarida et al., 2000). However, the smoothed regime probabilities are virtually identical (correlation 0.998), confirming that the regime dating is robust. The baseline specification is retained on the grounds that lagged realized inflation reflects the CBRT's actual information set.

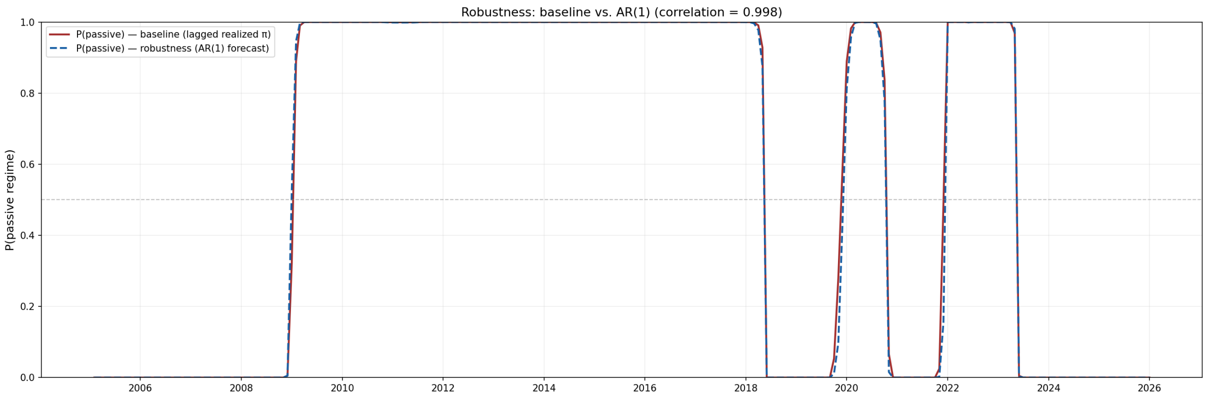

Figure 2: Robustness: regime probability correlation of 0.998 between lagged realized and AR(1) forecast inflation specifications.

5 Sectoral Inertia via TVP-Kalman Filtering

This section estimates the time-varying backward-indexation weight $\alpha_{i,t}$ for each CPI sub-index, classifies sub-indices into high- and low-inertia groups, and extracts common factors via PCA.

5.1 Model Specification

For each of the 42 three-digit CPI sub-indices, a state-space model is estimated:

Measurement equation:

$$\pi_{i,t} = \alpha_{i,t} \cdot \pi_{i,t-1} + \beta_i' X_t + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, \sigma^2_{\varepsilon,i}) \tag{5}$$

State equation:

$$\alpha_{i,t} = \alpha_{i,t-1} + \eta_{i,t}, \quad \eta_{i,t} \sim N(0, \sigma^2_{\eta,i}) \tag{6}$$

where $\pi_{i,t}$ is month-on-month inflation for sub-index i, X_t is a vector of controls (exchange rate change, commodity price change, output gap), and $\alpha_{i,t}$ is the time-varying backward-indexation weight. The hyperparameters $\sigma^2_{\varepsilon,i}$ and $\sigma^2_{\eta,i}$ are estimated by maximum likelihood for each sub-index individually, and the Kalman smoother produces smoothed estimates conditional on the full sample.

A high α indicates strong pass-through from lagged to current inflation. We label this *backward indexation* because contractual mechanisms (rental agreements linked to past CPI, service contracts with annual inflation adjustments) are a leading interpretation for the high-persistence sectors in the Turkish context. A low α indicates pricing driven primarily by current market conditions, characteristic of competitive sectors with frequent repricing. The parameter itself, however, identifies inflation *persistence*, not its source: adaptive expectations, sticky information, and rule-of-thumb pricing would generate a similar loading on lagged inflation. The label should accordingly be read as an interpretation rather than as an identified structural mechanism.

5.2 Estimation Results and Boundary Behavior

Of the 42 sub-indices, 10 produce $\sigma^2_{\eta,i}$ converging to zero, implying constant $\alpha_{i,t}$ across the sample. These include social protection ($\alpha = 0.485$), footwear (0.431), clothing (0.418), accommodation services (0.407), hospital services (0.375), package holidays (0.237), tobacco (0.223), alcoholic beverages (0.183), tertiary education (0.129), and other services (0.051). The boundary convergence likely reflects insufficient signal-to-noise ratio rather than genuinely fixed pricing behavior. These sub-indices are retained with their constant values; the re-embedding pattern is driven by the remaining 32 sub-indices with genuine time variation, whose $\sigma^2_{\eta,i}$ ranges up to 2.37.

5.3 Classification

Sub-indices are classified based on their post-2008 average $\alpha_{i,t}$. The cross-sectional mean across the 42 sub-indices is 0.450. Sub-indices at or above the mean are assigned to the **high-inertia group**; those below to the **low-inertia group**. This yields 19 high-inertia sub-indices (62.2% of the CPI basket) and 23 low-inertia sub-indices (37.6%).

Figure 3 visualizes the classification against CPI basket weight. Sub-indices in the upper-right quadrant drive the aggregate trajectory: actual rentals ($\alpha = 1.045$, weight 7.1%), catering (0.991, 8.3%), food (0.465, 23.5%), and transport services (0.478, 8.0%) together account for nearly half the CPI basket and exhibit substantial backward-indexation. The low-inertia group includes clothing (0.418, 6.9%), vehicles (0.337, 5.4%), and furniture (0.231, 3.2%), sectors where prices are determined by competitive dynamics and exchange rate pass-through.

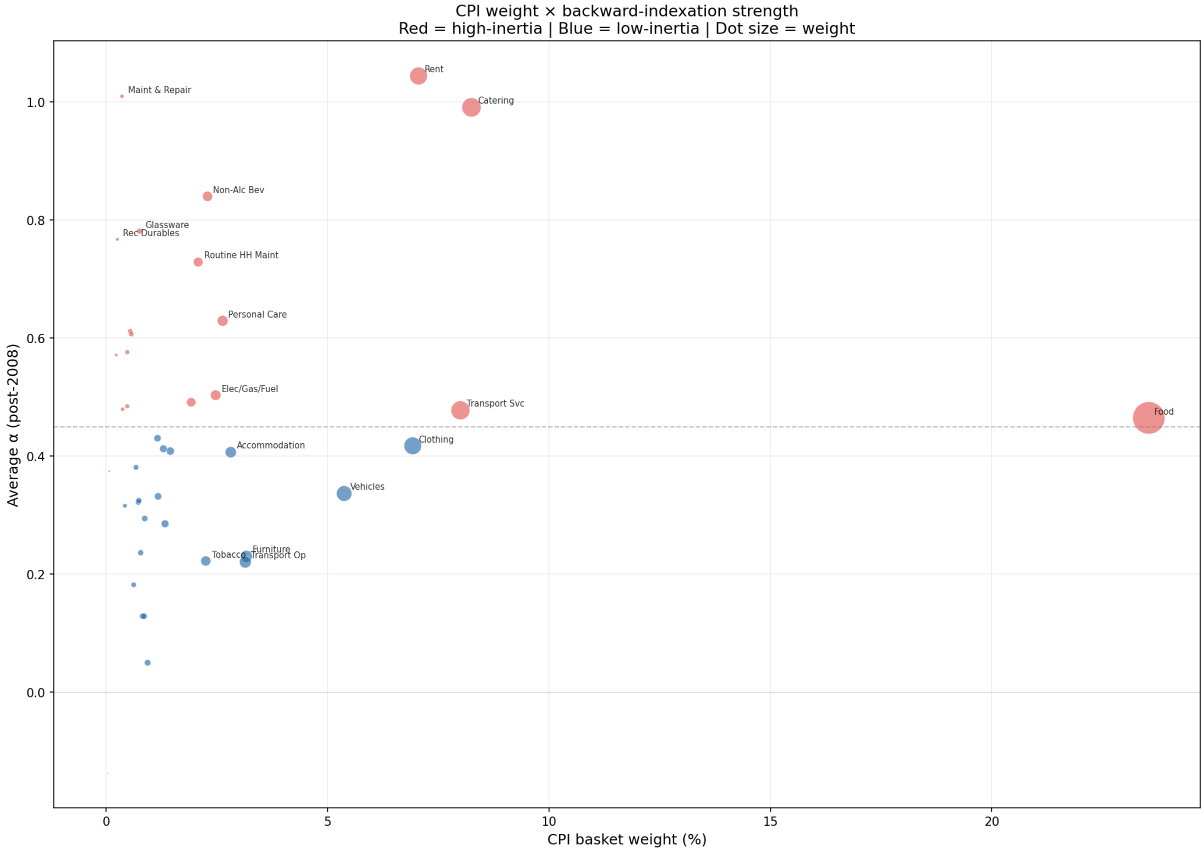

Figure 3: CPI basket weight versus average backward-indexation strength (α) for each sub-index. Red: high-inertia group. Blue: low-inertia group. Dot size proportional to CPI weight. Horizontal dashed line marks the mean classification threshold (0.450).

5.4 Aggregate Backward-Indexation Dynamics

Figure 4 plots the CPI-weighted aggregate backward-indexation intensity:

$$\bar{\alpha}_t = \frac{\sum_{i=1}^{42} w_i \cdot \alpha_{i,t}}{\sum_{i=1}^{42} w_i} \tag{7}$$

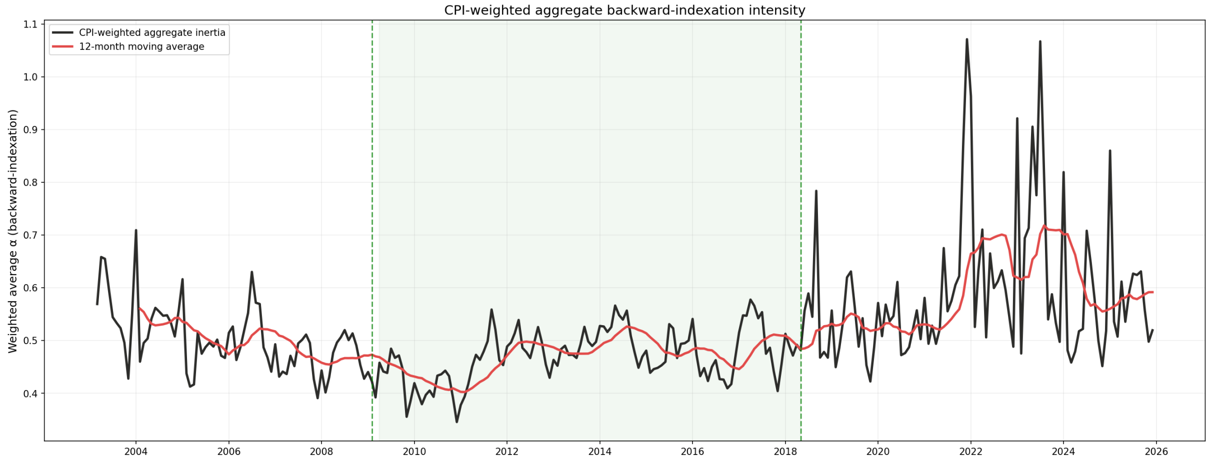

Figure 4: CPI-weighted aggregate backward-indexation intensity ($\bar{\alpha}_t$) with 12-month moving average. Green shading indicates the primary passive regime.

The aggregate backward-indexation weight declines from 0.51 during 2003–2007 to 0.44 during 2008–2010, consistent with the post-2001 reforms reducing contractual indexation as inflation fell to single digits. It reaches its trough around 2010–2011 and then rebuilds, gradually at first, rising only to 0.49 by 2015–2018, before accelerating to 0.58 in 2019–2022 and 0.62 by 2023–2025. The re-embedding thus *begins* during the primary passive regime identified in Section 4 but does most of its work afterward: the steepest increases occur after the episode ends in 2018. We document this co-movement without taking a stand on causation. The data establish that contractual persistence rebuilt over the same broad period in which monetary policy was accommodative; they do not, on their own, establish that the former was caused by the latter, which would require identification we do not pursue here. Table 3 reports the period averages by group.

Table 3: CPI-Weighted Backward-Indexation by Period and Group

Period	Aggregate	High-Inertia	Low-Inertia	Gap
2003–2007	0.511	0.633	0.308	0.325
2008–2010	0.437	0.569	0.218	0.351
2011–2014	0.488	0.599	0.304	0.295
2015–2018	0.491	0.603	0.304	0.299
2019–2022	0.576	0.703	0.366	0.337
2023–2025	0.619	0.767	0.376	0.391

Note: Weighted by 2026 CPI basket weights. Gap = High − Low.

Figure 5 decomposes the trajectory by group, confirming that the high-inertia group drives the aggregate dynamics.

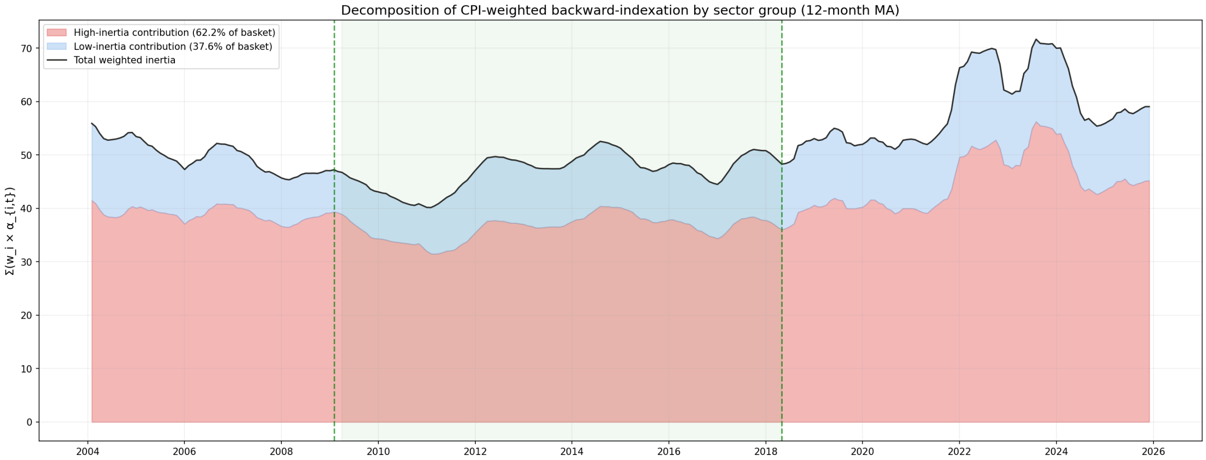

Figure 5: Decomposition of CPI-weighted backward-indexation by sector group (12-month moving average). Green shading indicates the primary passive regime.

6 The Cost of the Passive Regime: A Structural Counterfactual

The preceding sections establish two facts: the central bank's systematic response to inflation fell below the Taylor-principle threshold for an extended episode, and a majority of the consumption basket exhibits substantial inflation inertia. This section combines them to ask a quantitative question. Had the central bank instead maintained an active, rule-based stance throughout the 2009–2018 passive episode, how much lower would inflation have been, and through which component of the basket would the difference have operated?

6.1 The structural VAR

We estimate a seven-variable Markov-switching structural VAR on the monthly sample, ordering the variables as the output gap, the change in the global commodity price index, the change in the nominal exchange rate, the imported factor, the inertial factor, headline CPI inflation, and the policy rate. The system is estimated separately by regime under the regime path implied by the Markov-switching Taylor rule of Section 4, with the active and passive regimes accounting for 133 and 140 months respectively. Reduced-form coefficients are drawn from a Bayesian posterior under a Minnesota-type prior.

Monetary policy shocks are identified by sign restrictions following Uhlig (2005): a contractionary shock raises the policy rate over the months following impact, with magnitude bounds imposed in the spirit of Kilian and Murphy (2012) to discard implausibly large rotations. Identification is deliberately confined to the monetary shock, the only structural disturbance the counterfactual perturbs, rather than to the full set of structural shocks. The sign restrictions are essential to what follows: the reduced-form system exhibits a price puzzle, and only the identified responses deliver the economically correct sign of the inflation response to a policy contraction. The accepted draws confirm this, with the contractionary shock lowering CPI inflation over the horizon in both regimes.

6.2 The counterfactual experiment

The counterfactual replaces the passive systematic rule with a standalone active Taylor rule and re-simulates the entire system. The rule sets the policy rate as

$$i_t^* \;=\; r^* + \pi_t^{\text{ann}} + a(\pi_t^{\text{ann}})\,(\pi_t^{\text{ann}} - \pi^*)\,, \tag{8}$$

where π_t^{ann} is trailing twelve-month counterfactual inflation, $\pi^* = 3\%$ is the target, and $r^* = 2\%$ is the neutral real rate, so the rule prescribes a neutral nominal rate of 5% when inflation is at target. The gap coefficient $a(\cdot)$ eases linearly from 1.0 when inflation is at or above 10% to 0 at the target, so that the total inflation response $1 + a$ eases from a conventionally aggressive value of 2.0 down to the Taylor-principle floor of unity as inflation converges to target. The rule responds to inflation only; the output-gap term is set to zero, because the contemporaneous output gap in the estimated policy equation is contaminated by within-month endogeneity, as discussed in Section 4, and a wrong-signed loading would distort the experiment. A first-order smoothing parameter ($\rho = 0.3$) is applied to the prescribed rate to reflect the gradualism of actual policy.

Three features distinguish the experiment from a naive impulse-response exercise. First, it is *fully endogenous*: counterfactual inflation is generated by the joint dynamics of the system operating under the rule in equation (8). Second, the counterfactual rate is a *standalone* path determined by the rule, not the actual rate plus a deviation; it therefore does not inherit the discretionary rate cuts of the passive period. Third, transmission runs through the *identified* (sign-restricted) impulse responses, the entire seven-variable system evolves jointly rather than inflation alone.

Implementation follows the conditional-projection logic of Sims and Zha (2006). For each posterior draw, we solve in each month for the monetary policy shock required to bring the policy rate to the level prescribed by equation (8), conditional on the rate already delivered by the propagation of earlier shocks; this incremental treatment accounts for policy-rate persistence. The shock is propagated to all seven variables through the identified responses, the rule is re-evaluated on the updated counterfactual inflation, and the system is rolled forward to the end of the episode. Of the accepted sign-restricted draws, 493 of 496 yield non-explosive paths; the remainder are discarded. The implied policy innovations are small on average, with a mean absolute size of about 0.42 standard deviations of the estimated historical monetary shock.

6.3 Results

Figure 6 shows the path of annualized counterfactual inflation against the realized series. Under a consistently active rule, inflation tracks well below the actual path throughout the episode, averaging about 4.5% against a realized average near 8.3%. The counterfactual delivers a substantial but *partial* disinflation: it does not return inflation to the 3% target within the window.

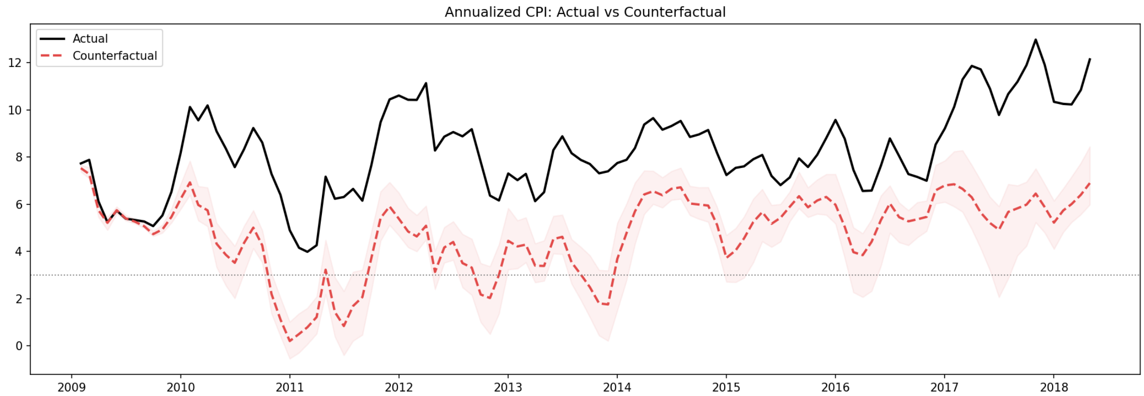

Figure 6: Annualized CPI inflation: actual versus counterfactual under a standalone active Taylor rule. The shaded band is the 68% posterior interval across sign-restricted draws.

Figure 7 shows the implied counterfactual policy rate against the realized series. The actual rate path has several distinct phases: a brief drop to around 1.5% from late 2010 to mid-2011 under the asymmetric corridor experiment, a return to 5% from August 2011 through 2012, a gradual ease to around 3.5% over 2013, and a sharp jump to around 8% with the January 2014 emergency hike, after which it settled near 7.5% for the rest of the episode. The counterfactual sits mostly above the realized rate but briefly fell below it in the months immediately after the August 2011 normalization and the January 2014 hike. The gap was positive in 82% of months, with a mean of 1.3 and median of 1.5 percentage points; the largest sustained gaps (around 2 pp on average) were during the asymmetric corridor period, 2013, and 2017–2018, and individual months reached above 4 pp. Sustained over nine years, this divergence produces the large cumulative inflation also reported below.

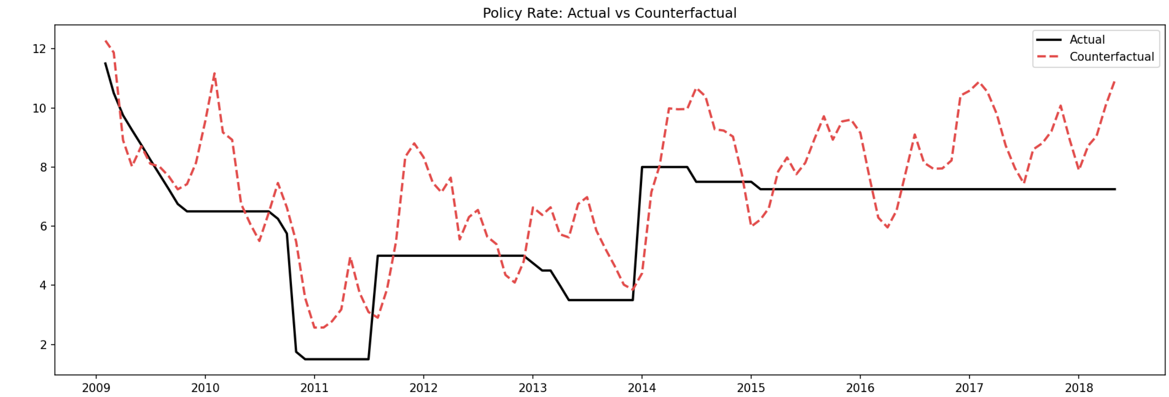

Figure 7: Policy rate: actual versus the standalone counterfactual rule. The counterfactual does not follow the discretionary cuts of 2011 and 2013.

The cumulative effect is summarized in Figure 8. By the end of the episode the price level is a median 34.5 percentage points lower under the active rule than under the realized passive policy, equivalent to an average gap of about 3.7 percentage points per year. The effect is sharply signed: the cumulative gap is positive in every posterior draw ($\Pr(\text{gap} > 0) = 1.00$), with a 68% credible interval of 27.9 to 40.4 percentage points and a 90% interval of 22.3 to 51.8.

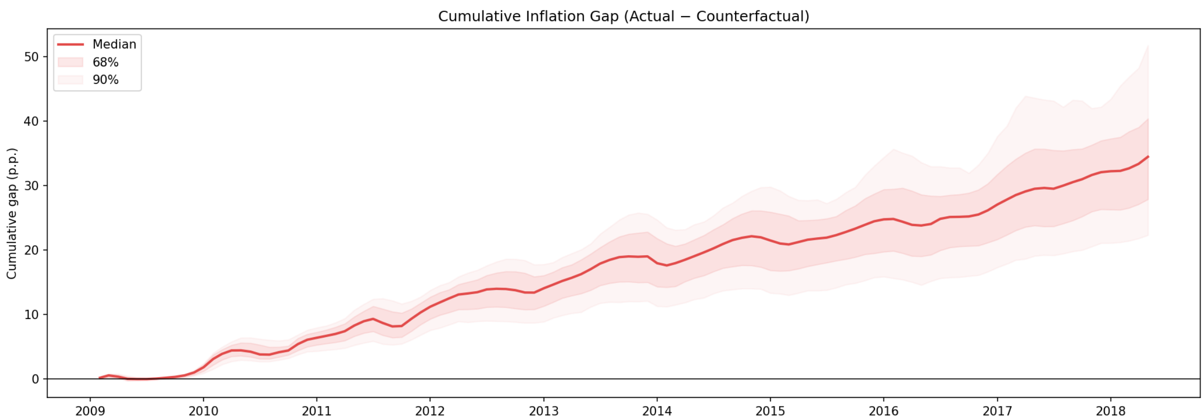

Figure 8: Cumulative inflation gap (actual minus counterfactual price level), with 68% and 90% posterior bands.

7 Conclusion

This paper has examined Turkey's inflation over the 2009–2018 passive monetary policy episode through three complementary analyses: the systematic stance of the central bank, the cross-sectional structure of inflation persistence, and a counterfactual quantification of what a consistently active rule would have delivered. A Markov-switching Taylor rule dates a passive policy regime in which the inflation response (0.67) fell well short of the Taylor-principle threshold, sustained for more than nine years. A time-varying-parameter filter shows that inflation inertia is pervasive, with sub-indices accounting for 62.2% of the basket lying above the cross-sectional mean of the persistence parameter, and a factor decomposition separates a dominant inertial component from a smaller imported one.

Bringing these together in a structural VAR, we construct a fully endogenous policy counterfactual: replacing the passive rule with a standalone active Taylor rule and re-simulating the whole system through identified monetary shocks. A consistently active stance would have lowered the cumulative price level by about 34.5 percentage points over the episode, roughly halving average inflation from 8.3% to 4.5%.

Several limitations should be acknowledged. The structural counterfactual rests on the assumption that the estimated transmission of monetary shocks is invariant to the change in the policy rule, a standard but non-trivial assumption when the regime shift is large. The analysis also does not decompose the residual gap between counterfactual inflation and target into channels: how much reflects the backward-looking pricing structure documented in Section 5 versus exchange-rate or commodity-price dynamics is left open. Finally, the counterfactual rule is calibrated to conventional benchmark values rather than estimated, and the posterior bands condition on the regime dates and latent factors as fixed inputs.

Several directions remain for future research. A sectoral extension of the counterfactual would identify where in the consumption basket the disinflation gap operates, sharpening the policy implications. Comparative applications to other emerging markets with documented episodes of weak monetary policy could test whether the magnitudes recovered here are characteristic of regime-change mechanisms more broadly or specific to Turkey. A regime-switching DSGE model would allow welfare-based comparison of the active and passive regimes, going beyond the orders-of-magnitude quantification provided here.

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A Supplementary Figures

This appendix collects the principal diagnostic figures supporting the main text. The counterfactual figures (counterfactual inflation, the policy-rate path, and the cumulative gap) appear inline in Section 6.

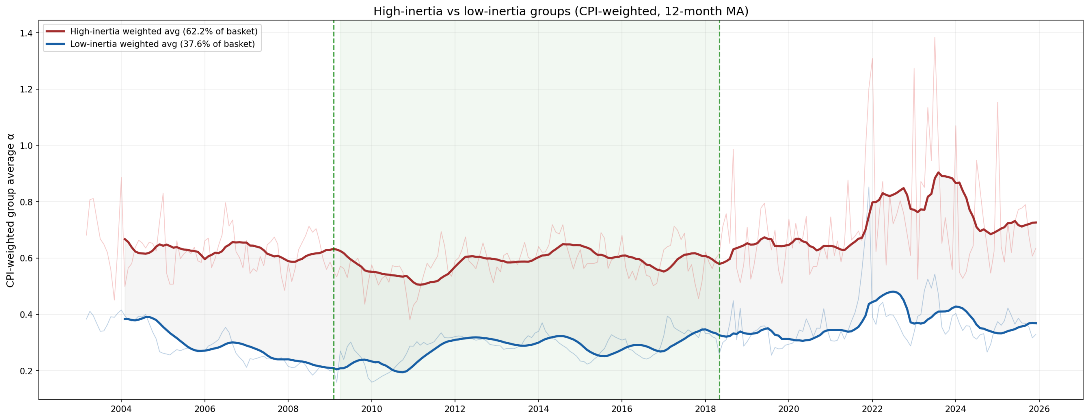

Figure 9: Weighted inflation in the high- and low-inertia groups, with sectors partitioned at the cross-sectional mean of the persistence parameter ($\bar{\alpha} = 0.450$): 19 sub-indices (62.2% of the basket) lie in the high-inertia group.

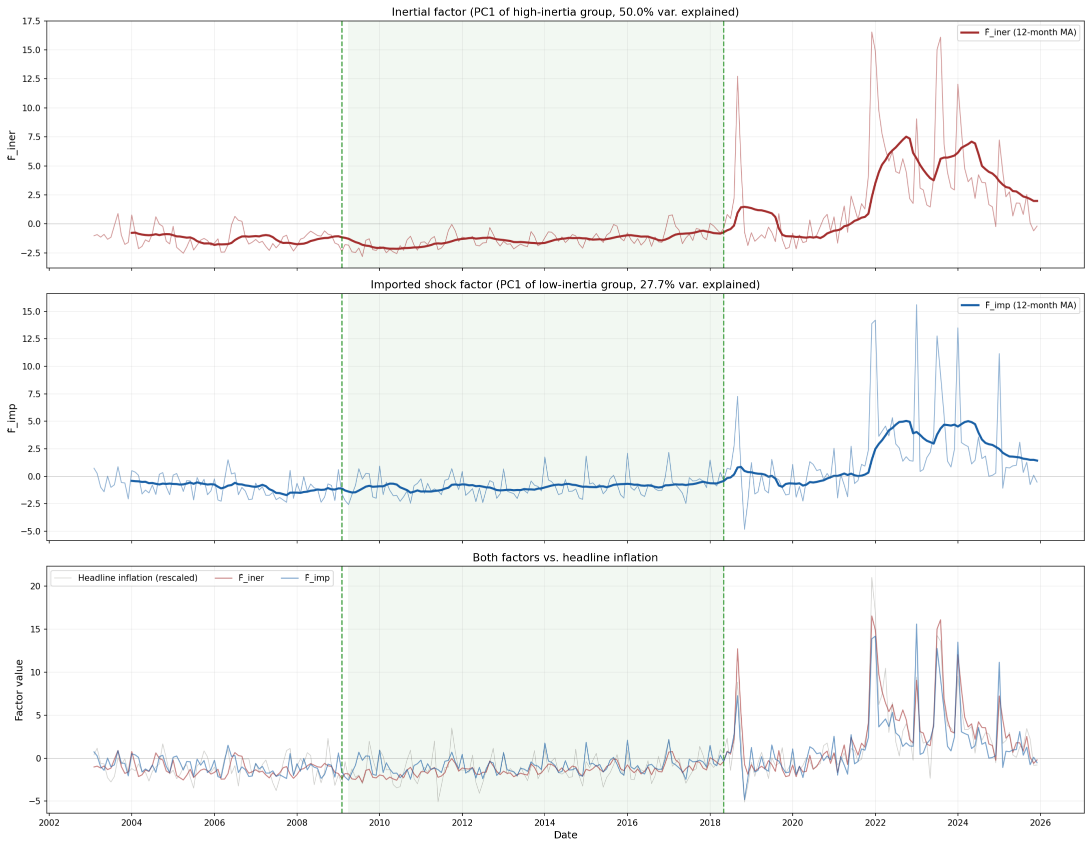

Figure 10: Extracted inertial factor (51% of variance) and imported factor (30% of variance) from the principal-components step (Section 5).

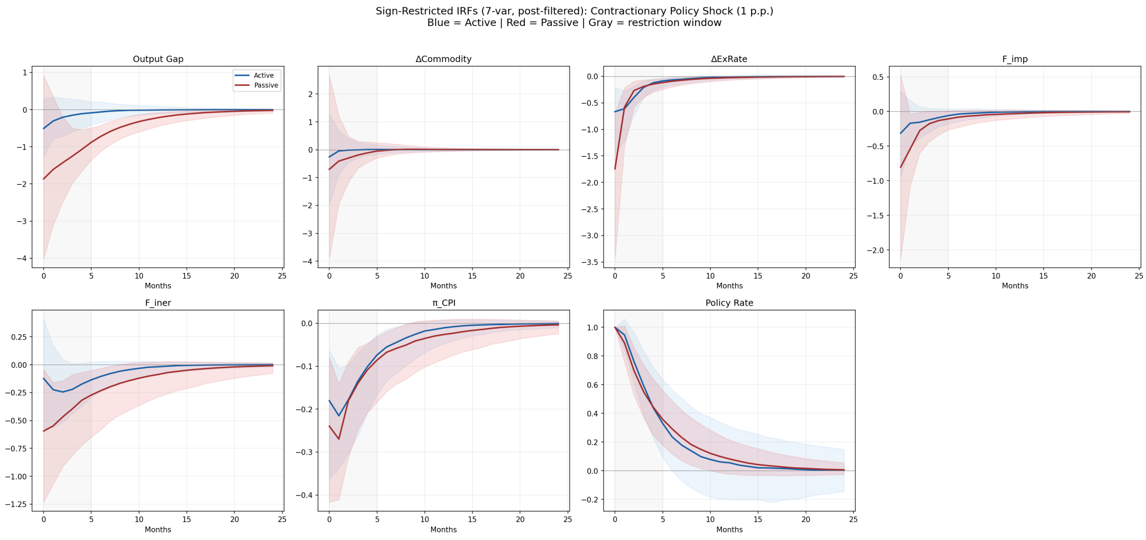

Figure 11: Sign-restricted impulse responses of the seven-variable system to an identified contractionary monetary policy shock, by regime; these responses underlie the counterfactual of Section 6.